

RESEARCH ARTICLE

High-Speed and Enhanced Motion Control for a Wheeled-Legged Humanoid Robot Using a Two-Wheeled Inverted Pendulum With Roll Joint

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ABSTRACT Wheeled-legged humanoid robots combine the mobility of wheels with the versatility of legs, offering significant advantages for locomotion. This paper proposes a hierarchical control framework for such robots, using a two-wheeled inverted pendulum with a roll joint (TWIP-R) as a template model. The framework integrates a motion planner and a whole-body controller. The motion planner utilizes a linear quadratic regulator (LQR) to dynamically adjust the zero moment point (ZMP), counteracting centrifugal forces and enabling stable, dynamic movements. Meanwhile, the whole-body controller (WBC), based on centroidal momentum, solves an optimization problem via quadratic programming (QP) while incorporating constraints from the rolling contact condition of the wheels and the dynamics of the humanoid robot. This framework generates optimized torque commands, enabling feasible and stable motion even in dynamic scenarios. Simulations featuring challenging maneuvers, such as slalom, demonstrate its ability to enhance stability and dynamic performance compared to traditional two-wheeled inverted pendulum (TWIP)-based methods by leveraging a dynamics model with roll motion. This framework demonstrates the potential of wheeled-legged humanoid robots to achieve dynamic, stable, and efficient locomotion in a variety of scenarios.

INDEX TERMS Optimal control, two-wheeled inverted pendulum with roll joint, wheeled-legged humanoid robot, whole-body control.

I. INTRODUCTION

Locomotion plays a significant role in fully maximizing the potential of humanoid robots [1]. However, most humanoid robots relying solely on legged locomotion face the challenge of slow walking speed. To address this, the integration of wheels with legs offers a promising solution, enhancing both performance and practicality. This hybrid approach provides several advantages, such as improved cost efficiency, adaptability to uneven terrain, and increased agility [2].

Recent studies have explored wheeled-legged robotic systems [3]. Traditionally, mobile robots have used legs

as a suspension system to assist in driving [4], [5]. While these approaches were effective on rough terrain, they did not provide adequate support for the active use of leg joints for complex movements such as walking or climbing stairs. Recent advances in wheel mounting on quadruped robots have enabled active use of leg joints for both walking and driving [6]. However, two wheeled-legged robots still face balance challenges due to constraints imposed by the zero moment point (ZMP) [7]. A representative example of overcoming these challenges to achieve high performance mobility is Boston Dynamics' Handle [8]. Other wheeled-legged robots with a closed-loop leg structure that demonstrate dynamic motion include Ascento [9] and Ollie [10]. In [11], a similar system was controlled

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using Cassie mounted on hovershoes. EvoBot [12] further showcases a unique design capable of performing locomotion and manipulation tasks simultaneously.

A. RELATED WORKS

The primary challenge for two-wheeled legged robots is maintaining balance during movement, particularly due to non-minimum phase dynamics, where movements oppose the direction of equilibrium during acceleration and deceleration phases [13]. Template models, such as the two-wheeled inverted pendulum (TWIP), simplify control by neglecting leg dynamics [14]. Approaches such as linear quadratic regulator (LQR) [15] and model predictive control (MPC) [16] have been applied to these systems, although MPC requires significant computational resources. More recently, reinforcement learning (RL) has been explored to address modeling errors and environmental variations [17].

Beyond balance control, researchers have investigated base height control for tasks such as jumping [18] and obstacle avoidance [19], often using whole-body control (WBC) to manage complex tasks [20]. WBC has been effectively applied to legged robots [21], [22] as well as wheeled-legged robots, including both two wheeled [23] and four wheeled-legged configurations [24]. Its generalized formulation serves as a versatile dynamic motion controller for a variety of systems and tasks.

In [25], the cart-linear inverted pendulum model (cart-LIPM) and under-actuated LIPM generated walking and driving motions but lacked the ability to perform curved driving. While some systems have used the roll motion of the base to stabilize the curved movement [26], the floating base alone cannot generate the required roll motion for wheeled-legged humanoid robots with high degree-of-freedom (DoF). In our previous work, we demonstrated both straight and curved driving using LIPM and a cart-table model, validated through optimal whole-body control [27]. However, challenges remained in coordinating control between two independent template models and generating rotational motion directly, preventing full utilization of roll motion.

To address these limitations, we introduce the two-wheeled inverted pendulum with roll joint (TWIP-R) template model, which incorporates pitch, yaw, and roll motions. Previous approaches, such as [28], added a roll-axis actuator but relied on independent lateral dynamics unrelated to the TWIP. Consequently, they were not sufficient to achieve comprehensive rotational dynamics. Similarly, [29] employed two ball screw actuators for the structural motion of the roll, which inherently limited the roll range. In contrast, TWIP-R simplifies the dynamics of wheeled-legged humanoid robots by directly integrating roll motion into the system's rotational dynamics. By adding a single revolute joint to the standard TWIP, the proposed model enables lateral rotation, resulting in a compact and effective system capable of pitch, yaw, and roll motions.

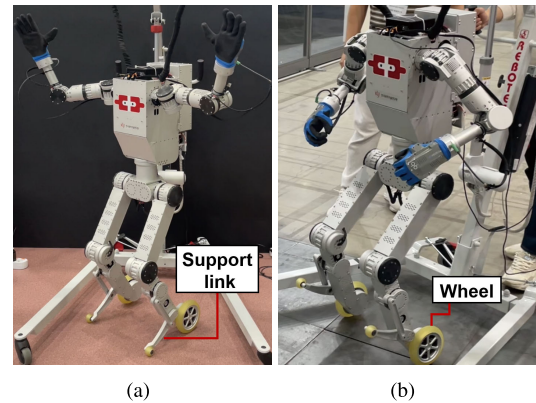


FIGURE 1. The real wheeled-legged humanoid robot with support links attached to the calf links. (a) The robot standing stably with support links lowered. (b) The robot in motion after raising the support links.

In our prior work [30], we addressed the nonlinear model of TWIP-R using model predictive path integral (MPPI). While MPPI provided a solution for TWIP-R, it encountered issues such as chattering in input acceleration and high computational demands due to random sampling, necessitating GPU-based computations. As an alternative, we adopted LQR for its stability and robustness in similar systems. Through partial feedback linearization (PFL) [31], [32], we retained the nonlinear characteristics of the TWIP-R system as much as possible while linearizing it into an acceleration input form. Consequently, TWIP-R provides reference values for WBC, allowing the entire robot to perform multiple tasks simultaneously while maintaining stable and dynamic driving locomotion. In particular, the roll motion demonstrates exceptional performance in curved trajectories.

B. CONTRIBUTION

We present a hierarchical WBC framework approach using the TWIP-R template model for the wheeled-legged humanoid robot, as shown in Fig. 1. The main contributions of this work are as follows:

- 1) Proposed a motion planner using a linearized TWIP-R model with LQR control, which overcomes the non-minimum phase dynamics while considering the dynamic characteristics introduced by the roll joint. This approach resolves limitations of the traditional TWIP model.
- 2) Proposed a centroidal momentum-based WBC designed to track feasible references generated by the TWIP-R model, while addressing the characteristics of wheeled-legged humanoid robots, such as nonholonomic constraints and the efficient management of multiple tasks.

The overall system is illustrated in Fig. 2, which consists of two main components: a template-based motion planner and an optimization-based WBC. In our approach, a virtual TWIP-R is controlled using LQR to determine the motions of the virtual mobile base and pole while maintaining balance. These motions are then transformed into feasible trajectories

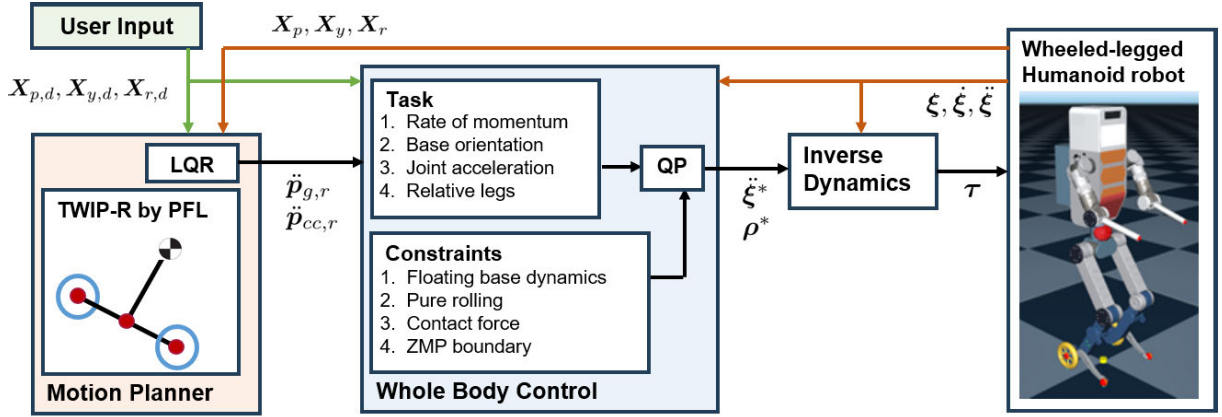


FIGURE 2. Overview of the suggested framework, divided into a motion planner and a controller. The motion planner receives user input and generates executable motion for the whole-body control. The whole-body controller operates the robot by calculating torque to track its motion.

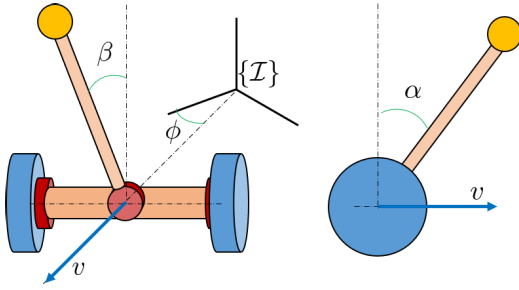


FIGURE 3. Schematic representation of the two-wheeled inverted pendulum with roll model. The virtual mobile base moves forward with velocity v and steers at an angle ϕ relative to the global frame. The virtual pole has a pitch angle α and a roll angle β .

for the wheels and center of mass (CoM), a process described in Section II. The WBC tracks the planned motions, enabling stable locomotion by handling multiple tasks with weighted priorities and respecting dynamic constraints. Optimized joint accelerations and contact forces are converted into actuator torques through inverse dynamics, as explained in Section III. The performance of the proposed method is validated in Section IV, with continuous state feedback provided through simulation.

II. LQR BASED MOTION PLANNER

The proposed motion planner generates motion based on user inputs, including velocity, angular velocity, pitch, roll angles of the virtual TWIP-R model. The virtual base is used to create the desired trajectory, while the pitch and roll are utilized to maintain balance. This section first discusses the linearization of the TWIP-R model using the PFL approach, followed by the determination of roll motion. Finally, the feedback control design, including the LQR controller based on the linearized model, is presented. The structure of the TWIP-R is illustrated in Fig. 3.

A. DECOUPLED TWIP-R SYSTEM USING PFL

The equation of motion of TWIP-R shown in (1) is derived using the Lagrangian method as follows:

$$M_o \ddot{\theta}_o + C_o \dot{\theta}_o + g_o = B_o^T \tau_o + J_o^T \lambda_o \quad (1)$$

where

$$\begin{aligned} \theta_o &= [x_m \ y_m \ \phi \ \alpha \ \theta_{rw} \ \theta_{lw} \ \beta]^T \\ \tau_o &= [\tau_{rw} \ \tau_{lw} \ \tau_\beta]^T \\ B_o &= \begin{bmatrix} 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \end{aligned}$$

and x_m, y_m and ϕ represent the x, y positions and the steering angle of the mobile base, respectively. α and β denote the pitch and roll angles of the pole, respectively. Similarly, θ_{rw} and θ_{lw} are the rotation angles of the right and left wheels, respectively.

Nonholonomic constraints for no slip can be expressed as follows:

$$J_o \dot{\theta}_o = 0, \quad (2)$$

where

$$J_o = \begin{bmatrix} -\sin \phi & \cos \phi & 0 & 0 & 0 & 0 & 0 \\ \cos \phi & \sin \phi & w_b & 0 & -r_w & 0 & 0 \\ \cos \phi & \sin \phi & -w_b & 0 & 0 & -r_w & 0 \end{bmatrix}.$$

To simplify the dynamics and remove Lagrange multipliers, we use the N_o matrix, which is the null space matrix of J_o .

$$\dot{\theta}_o = N_o \dot{v}, \quad (3)$$

where

$$N_o = \begin{bmatrix} \cos \phi & 0 & 0 & 0 \\ \sin \phi & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1/r_w & 0 & w_b/r_w & 0 \\ 1/r_w & 0 & -w_b/r_w & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \dot{v} = \begin{bmatrix} \dot{v}_m \\ \dot{\alpha} \\ \dot{\phi} \\ \dot{\beta} \end{bmatrix},$$

where w_b is the distance from the base to the wheel, and r_w is the radius of the wheel. The reduced equation of motion is then derived as follows:

$$M_t \ddot{v} + C_t \dot{v} + g_t = M_t \dot{v} + h_t = B_t^T \tau_t. \quad (4)$$

The TWIP with the added roll joint shows complex nonlinearities, which require linearization to enable LQR control. First, the joint torque inputs in (4) are transformed into motion accelerations through the structural relationship in (5).

$$\begin{bmatrix} 1/r_w & 1/r_w & 0 \\ -1 & -1 & 0 \\ w_b/r_w & -w_b/r_w & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tau_{rw} \\ \tau_{lw} \\ \tau_\beta \end{bmatrix} = \begin{bmatrix} u_P \\ -r_w u_P \\ u_Y \\ u_R \end{bmatrix}. \quad (5)$$

The dependency between the first and second rows of (5) can be observed. By combining them, (6) can be formulated as follows:

$$(r_w m_{11} + m_{12})\ddot{v}_m + (r_w m_{12} + m_{22})\ddot{\alpha} + (r_w m_{13} + m_{23})\ddot{\phi} + r_w m_{14}\ddot{\beta} + (r_w h_1 + h_2) = 0, \quad (6)$$

where m represents an element of the inertia matrix, with subscripts indicating the row and column numbers. h represents an element of the nonlinear vector, with the subscript indicating the row number.

The zero dynamics equation for $\ddot{\alpha}$ obtained from (6) is substituted into the reduced equation of motion. Using PFL, the reduced equation of motion is partially linearized with the new input, resulting in the derivation of three state-space equations for the pitch, yaw, and roll systems.

$$\dot{X}_p = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ a_{20} & 0 & 0 \end{bmatrix} X_p + \begin{bmatrix} 0 \\ 1 \\ b_2 \end{bmatrix} a_p, \quad (7)$$

$$\dot{X}_y = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X_y + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a_y, \quad (8)$$

$$\dot{X}_r = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X_r + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a_r, \quad (9)$$

where

$$\begin{aligned} X_p &= \begin{bmatrix} \alpha \\ v_m \\ \dot{\alpha} \end{bmatrix}, \quad X_y = \begin{bmatrix} \phi \\ \dot{\phi} \end{bmatrix}, \quad X_r = \begin{bmatrix} \beta \\ \dot{\beta} \end{bmatrix}, \\ a_p &= \ddot{v}_m, \quad a_y = \ddot{\phi}, \quad a_r = \ddot{\beta}, \\ p_1 &= m_w + \frac{J_{wa}}{r_w^2}, \quad p_2 = I_{p_y} + m_p l_g^2, \\ a_{20} &= \frac{\bar{g} m_p l_g}{I_{b_y} + m_p l_g r_w + p_2}, \\ b_2 &= -\frac{m_p l_g + r_w(m_p + m_b + 2p_1)}{I_{b_y} + m_p l_g r_w + p_2}, \end{aligned}$$

where m_w , m_p and m_b represent the mass of wheel, pole, and base, respectively. $\bar{g}$ is the gravitational acceleration, and l_g is the distance from the base to the center of mass. I_{p_y} , I_{b_y} are the moment of inertia about the y-axis of the pole and base, respectively. And J_{wa} represents the moment of inertia about the wheel rotation axis.

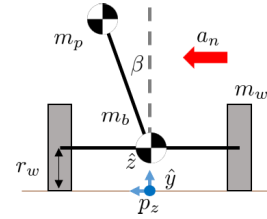


FIGURE 4. Schematic diagram illustrating the roll reference value to shift the ZMP for counteracting centrifugal force.

B. ROLL REFERENCE

Typically, pitch motion is controlled by setting the desired values α_d and $\dot{\alpha}_d$ to zero. This ensures balance by avoiding unnecessary tilting. In contrast, roll motion requires non-zero desired values, β_d , to counteract centrifugal forces that arise during dynamic movements, as illustrated in Fig. 4. Roll motion is actively generated and controlled using the ZMP equation, which accounts for the effects of centrifugal acceleration [33].

$$p_{z,y} = \frac{-(m_b + 2m_w)a_n r_w - m_p \bar{g} l_g \beta - m_p a_n p_{bg,z}}{(m_b + 2m_w)\bar{g}}, \quad (10)$$

where a_n denotes centrifugal acceleration, $p_{bg,z}$ is z position from base to CoM.

By setting $p_{z,y} = 0$, the roll motion required to shift the ZMP to the center of the most stable support line on a curved trajectory, where centrifugal forces arise, can be derived as follows:

$$\beta_d = \frac{[m_p p_{bg,z} + (m_b + 2m_w)r_w] a_n}{m_p \bar{g} l_g}. \quad (11)$$

C. LQR FEEDBACK CONTROL

The three decoupled systems are solved using LQR, which controls the pole's tilting and leaning angles to maintain balance. Additionally, the velocity and steering angle of mobile base are controlled to track a given motion. The optimization equation for the LQR, which minimizes the quadratic cost function is shown in (12).

$$J_j = \int_0^\infty X_j^T Q_j X_j + a_j^T R_j a_j, \quad \text{where } j \in p, y, r, \quad (12)$$

where Q , R are weight matrices such that $Q = Q^T \geq 0$, $R = R^T > 0$. Three equations are formulated by defining each cost function, state vector, weight matrix, and input vector using the state-space representations of pitch, yaw, and roll systems. Subsequently, Equation (7), (8), and (9) are solved by the Continuous Algebraic Riccati Equations (CARE) to obtain the gains K_p , K_y , K_r . The corresponding feedback control is expressed in (13).

$$a_j = -K_j(X_{j,d} - X_j), \quad \text{where } j \in p, y, r. \quad (13)$$

This process allows for computing the accelerations required to track the desired trajectory. By substituting these

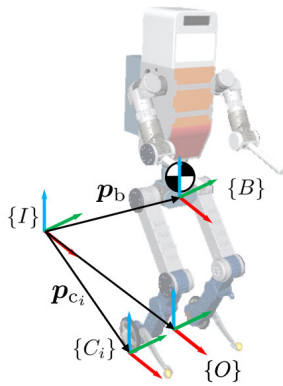


FIGURE 5. Visualization of the coordinate frames and key reference points used for modeling the robot's dynamics. The relationships between the robot's base, contact points, and inertial frame are depicted for clarity.

accelerations into the state-space equations (7), (8), (9), the desired position, velocity, and acceleration for each state can be obtained. Using this information, the kinematics of the TWIP-R can be computed to determine the motion of the virtual mobile base and the poles' tip, which serve as the center of the contact points and the CoM that the humanoid robot should track.

III. CENTROIDAL MOMENTUM-BASED WHOLE-BODY CONTROL

The whole-body control uses the reference motion generated by the motion planner to compute optimal control solutions through Quadratic Programming (QP). By incorporating physical and dynamic constraints while minimizing a quadratic cost function, WBC determines the optimal joint accelerations and contact forces required to execute multiple tasks. Weight matrices prioritize these tasks, ensuring the simultaneous execution of objectives such as balance and trajectory tracking.

A. MODELING

Accurate modeling of the wheeled-legged humanoid robot for effective WBC. This section describes the coordinate frames and dynamics equations used in the system.

1) COORDINATES AND FRAMES

The generalized coordinates of the robot, including velocities and accelerations, are defined as follows:

$$\dot{\xi} = \begin{bmatrix} \dot{p}_b \\ \omega_b \\ \dot{q} \end{bmatrix}, \quad (14)$$

where $\dot{\xi} \in \mathbb{R}^{6+n}$ represents the velocities of the generalized coordinates, and n denotes the number of joints. $\dot{p}_b$, ω_b are the linear and angular velocities of the floating base with respect to the inertial frame, and $\ddot{p}_b$, $\dot{\omega}_b$ indicate the linear and angular accelerations of floating base, respectively. Additionally, $\dot{q}$, $\ddot{q}$ refer to the joint angular velocities and accelerations.

The coordinate frames used for modeling and control are shown in Fig. 5. The frames $\{I\}$, $\{B\}$, and $\{G\}$ represent the

inertial frame, floating base frame, and center of mass frame, respectively. Additionally, $\{O\}$ represents an orientation frame that matches only the yaw angle of $\{B\}$ on the frame of $\{I\}$. The frame $\{C\}$ denotes the contact frames corresponding to the points where the wheels make contact with the ground.

2) ROBOT DYNAMICS EQUATIONS OF MOTION

The dynamics of wheeled-legged humanoid robot is expressed as follows:

$$\begin{bmatrix} M_b \\ M_q \end{bmatrix} \ddot{\xi} + \begin{bmatrix} C_b \\ C_q \end{bmatrix} \dot{\xi} + \begin{bmatrix} g_b \\ g_q \end{bmatrix} = \begin{bmatrix} 0_{6 \times 1} \\ \tau \end{bmatrix} + \sum_k^{n_c} \begin{bmatrix} X_{c_k}^T \\ J_{c_k}^T \end{bmatrix} f_{c_k}, \quad (15)$$

where

$$X_{c_k}^T = \begin{bmatrix} I_{3 \times 3} \\ [p_{bc_k}]_{\times} \end{bmatrix}, \quad k = r, l,$$

and $M, C \in \mathbb{R}^{(6+n) \times (6+n)}$ and $g \in \mathbb{R}^{(6+n)}$ represent the inertia, coriolis and centrifugal matrix, and gravitational force vector, respectively, which can be decomposed into the floating base and joint parts. τ denotes the joint torque, and X_c, J_c denote the base and joint part of the contact position jacobian, respectively. f_c represents the contact force. n_c is the number of contacts.

B. QP FORMULATION FOR WBC

The WBC problem is formulated as a QP optimization as follows:

$$\begin{aligned} \min_{\ddot{\xi}, \rho} \quad & \sum_{m=1}^{n_t} W_m \| (A_m z - b_m) \|^2 \\ \text{subject to} \quad & C_{ec} z = d_{ec} \\ & C_{ic} z \leq d_{ic}, \end{aligned} \quad (16)$$

where

$$z = \begin{bmatrix} \ddot{\xi}^T & \rho^T \end{bmatrix}^T,$$

where A_m, b_m represent the task matrix and vector for the m -th task, respectively. C_{ec}, d_{ec} denote equality constraints, and C_{ic}, d_{ic} represent inequality constraints. W_m denotes the weight for each task, and z is the optimization variable consisting of joint accelerations and contact force magnitudes.

C. TASKS FOR WHEELED LOCOMOTION

The cost function in WBC includes several tasks, with a regularization term added to optimization stability. While many tasks are similar to those in legged robot control [21], certain tasks are tailored to leverage the capabilities of wheeled locomotion.

1) RATE OF CENTROIDAL MOMENTUM FOR COM MOTION

The centroidal momentum of the CoM represents the overall motion of the entire robot [34]. Instead of the commonly used LIPM for walking locomotion, the motion of the

pole's end from the TWIP-R model, combined with a user-defined height, is used to control the rate of centroidal linear momentum. The rate of centroidal angular momentum is not directly controlled; instead, the overall angular balance of the robot is managed through the orientation of the floating base and the wheel movements of the TWIP-R.

$$A_1 = [M_b \quad 0], \quad b_1 = \begin{bmatrix} X_{bg}^T \dot{h}_{g,r} - C_b \dot{\xi} \\ 0 \end{bmatrix}, \quad (17)$$

where $\dot{h}_{g,r}$ represents the rate of momentum reference. The linear term of $\dot{h}_{g,r}$ takes the acceleration reference of CoM, ensuring that the robot's CoM trajectory matches the motion of the virtual TWIP-R's pole tip.

$$\ddot{p}_{g,r} = \ddot{p}_{g,d} + K_P(p_{g,d} - p_g) + K_D(\dot{p}_{g,d} - \dot{p}_g), \quad (18)$$

where $p_{g,d}$, $\dot{p}_{g,d}$, and $\ddot{p}_{g,d}$ represent the desired position, velocity, and acceleration of the CoM, respectively. K_P and K_D denote the proportional and derivative gains used to calculate the CoM acceleration reference.

2) ORIENTATION OF FLOATING BASE FOR POSTURE

The orientation of the floating base determines the posture of the robot. Notably, the base orientation and the overall balance of the robot are distinct concepts.

$$A_2 = [I \quad 0], \quad b_2 = \begin{bmatrix} \dot{\omega}_b \\ 0 \end{bmatrix}, \quad (19)$$

where I is identity matrix.

3) JOINT ACCELERATION FOR DRIVING

The robot's joints are divided into three parts: upper body joints (waist and arms), lower body joints, and wheel joints. For stable and high-speed driving, minimizing unnecessary movement in the upper body is prioritized.

$$A_3 = [I \quad 0], \quad b_3 = \begin{bmatrix} \ddot{q}_r \\ 0 \end{bmatrix}, \quad (20)$$

Unlike walking locomotion, wheeled-legged humanoid robots primarily move forward due to the nonholonomic constraints of the wheels, which limit lateral motion. Wheel joint accelerations are critical for balance and trajectory tracking. The reference acceleration for each wheel is derived based on the center point of the two contact points.

$$\begin{aligned} \ddot{q}_{rw,r} &= (\ddot{x}_{cc,r} + w_b \ddot{\phi}_r) / r_w \\ \ddot{q}_{lw,r} &= (\ddot{x}_{cc,r} - w_b \ddot{\phi}_r) / r_w \end{aligned} \quad (21)$$

where $\ddot{x}_{cc,r}$ and $\ddot{\phi}_r$ are obtained from the forward motion of the virtual mobile base in the TWIP-R model. Unlike the template model, the wheeled-legged humanoid robot tilts its wheels due to roll motion, resulting in lateral movement of the center point between the wheels. To address this issue, the generated motion is projected onto the center point between the two contact points.

$$\ddot{x}_{cc,r} = \ddot{x}_{cc,d} + K_{P_{cc}}(x_{cc,d} - x_{cc}) + K_{D_{cc}}(\dot{x}_{cc,d} - \dot{x}_{cc}), \quad (22)$$

$$\ddot{\phi}_r = \ddot{\phi}_d + K_{P_\phi}(\phi_d - \phi) + K_{D_\phi}(\dot{\phi}_d - \dot{\phi}), \quad (23)$$

4) RELATIVE LEGS FOR BALANCE

Since wheels are actively utilized for locomotion, additional leg movements are possible. However, uncontrolled leg movements can cause imbalance. To prevent this, the relative positions of the legs are actively controlled to maintain alignment, ensuring stability by minimizing leg divergence. In this formulation, ${}^o J_{p,e}$ represents the jacobian matrix of the end effector, ${}^o \dot{J}_{p,e}$ denotes the time derivative of the jacobian matrix and ${}^o \ddot{p}_{e,r}$ is the reference acceleration of the end effector. These parameters are used to compute the desired motion of the legs in operational space.

$$A_4 = [{}^o J_{p,e} \ddot{\xi} \quad 0], \quad b_4 = \begin{bmatrix} {}^o \ddot{p}_{e,r} - {}^o \dot{J}_{p,e} \dot{\xi} \\ 0 \end{bmatrix}, \quad (24)$$

D. CONSTRAINTS FOR WHEELED-LEGGED HUMANOID ROBOTS

To ensure stability and adhere to the physical principles governing wheeled-legged humanoid robots, we imposed a set of equality and inequality constraints.

1) FLOATING BASE DYNAMICS

The dynamics of the floating base, derived from physical principles, are expressed as equality constraints. These equations, presented in (15), are reformulated in terms of force magnitudes using the basis matrix of the modeled contact forces.

$$M_b \ddot{\xi} + C_b \dot{\xi} + g_b = X_c^T U \rho, \quad (25)$$

where U and ρ are the merged basis matrix and the merged force magnitude vector.

2) PURE ROLLING CONTACT

We assume point contact between the flat ground and the wheels. To maintain pure rolling motion, the relative velocity and acceleration between the wheel and the ground must be zero. Although both involve point contact, there is a significant difference compared to bipedal robots without wheels. Rolling contact introduces an additional instantaneous velocity term due to the interaction between the wheel and the ground, which not present in standard rigid-body motion [35], [36]. The relative velocity and acceleration condition are expressed as:

$$\begin{aligned} \dot{p}_{rel} &= \dot{p}_{c_w} - \dot{p}_{c_e} \\ &= \dot{p}_w + \omega_w \times R_w^w p_{wc} + \underbrace{R_w^w \dot{p}_{wc}}_{\text{rolling contact term}} = 0, \end{aligned} \quad (26)$$

$$\begin{aligned} \ddot{p}_{rel} &= \ddot{p}_{c_w} - \ddot{p}_{c_e} \\ &= \ddot{p}_w + \dot{\omega}_w \times R_w^w p_{wc} + \omega_w \times (\omega_w \times R_w^w p_{wc}) \\ &\quad + \underbrace{\omega_w \times R_w^w \dot{p}_{wc}}_{\text{rolling contact term}} = 0, \end{aligned} \quad (27)$$

where the $\dot{p}_{rel}$, $\ddot{p}_{rel}$ represent the relative velocity and acceleration between the contact point on the wheel (c_w) and the ground (c_e). Pure rolling motion is ensured by optimizing

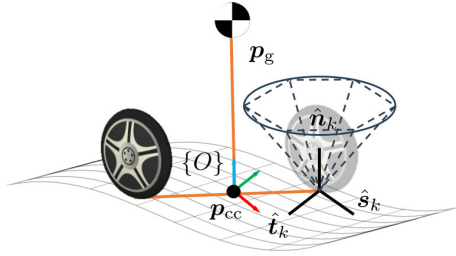


FIGURE 6. Illustration of the friction cone demonstrating the interaction between the wheel and the ground. The midpoint between the two contact points is used as a reference for the virtual mobile base. Here, $\hat{n}$, $\hat{s}$, and $\hat{t}$ represent the contact frame.

acceleration equality constraints that reflect these rolling conditions.

$${}^c J_c \ddot{\xi} + {}^c \dot{J}_c \dot{\xi} + R_c^T (\omega \times R_w^w \dot{p}_{wc}) = 0, \quad (28)$$

where ${}^c J_c$, ${}^c \dot{J}_c$ denotes the jacobian of the contact point and its time derivative with respect to contact frame.

3) FRICTION CONE APPROXIMATION

To prevent slipping, a unilateral inequality constraint is applied to ensure that the contact forces remain within the limits of the friction cone, as shown below:

$$\rho \geq 0. \quad (29)$$

The friction cone is modeled to Coulomb friction and approximated as a 6-dimensional polygon for computational efficiency [37], as shown in Fig. 6. The contact forces are expressed as follows:

$$f_{c_k} = U_k \rho_k = [u_{k,1} \cdots u_{k,m}] \begin{bmatrix} \rho_1 \\ \vdots \\ \rho_m \end{bmatrix}, \quad (30)$$

where

$$u_{k,i} = \begin{bmatrix} C_f \cos \frac{2\phi(i-1)}{m} \\ C_f \sin \frac{2\phi(i-1)}{m} \\ 1 \end{bmatrix}, \quad m = 6$$

where U_k and ρ_k denote basis vectors and a force magnitude vector, respectively, and C_f represents the coefficient of friction. This approximation enhances computational efficiency without significantly compromising the accuracy of contact force modeling.

4) BOUNDARY OF ZMP SUPPORT LINE

Unlike walking robots with feet, which have a ZMP support area defined by the surface of their feet, wheeled-legged humanoid robots have a ZMP support line defined by the two wheel contact points. The lateral position of the ZMP is therefore critical for maintaining balance. To ensure stability, the ZMP is constrained to remain within the ZMP support line, as follows:

$$l_{min} \leq {}^o y_z - {}^o y_{cc} \leq l_{max} \quad (31)$$

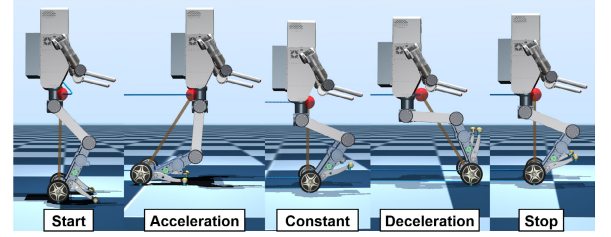


FIGURE 7. Snapshot of each section of straight trajectory. Our framework-controlled wheeled-legged humanoid robot actively utilizes its legs to adapt to the characteristics of each section.

where l_{min} , l_{max} represent the minimum and maximum bounds, ${}^o y_z$ denotes the y position of ZMP, and ${}^o y_{cc}$ represents the y position of the center point between the two contacts.

E. TORQUE COMMAND WITH INVERSE DYNAMICS

The torques are computed by substituting the optimized values of $\ddot{\xi}$ and ρ into the dynamic equation, considering only the actuator components of (15).

$$\tau = M_q \ddot{\xi}^* + C_q \dot{\xi} + g_q - \sum_k^{n_c} J_{c_k}^T f_{c_k}^*, \quad (32)$$

where $f_{c_k}^* = U_k \rho_k^*$, $k \in n_c$.

IV. SIMULATION

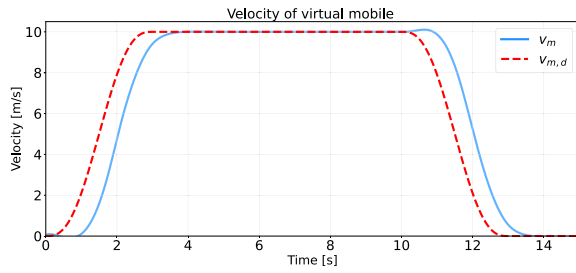
This section describes simulation experiments conducted to evaluate the control performance of our wheeled-legged humanoid robot. The simulations were performed using the MuJoCo physics engine [38]. Dynamic modeling, including the solution of the CARE for LQR, was implemented using our custom-developed library. The QP problem was solved using Quadprog++ [39], and the overall control framework algorithm was implemented in C++.

A. TRACKING PERFORMANCE

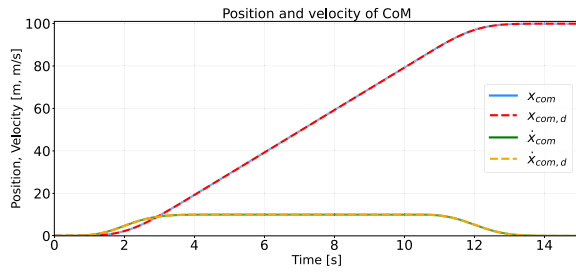
For a straight trajectory, the robot accelerates to 10 m/s over 3 seconds, followed by a constant velocity phase before decelerating to a stop. The articulated joints in our model provide increased flexibility in controlling the CoM, improving stability across a wider range. As illustrated in Fig. 7, the proposed framework effectively utilizes leg movements to adapt to various phases of the trajectory. In Fig. 8(a), the control method demonstrates precise tracking of user-defined inputs, with minimal overshoot during acceleration and minimal undershoot during deceleration. Furthermore, Fig. 8(b) shows that WBC ensures accurate CoM motion tracking, maintaining balance throughout the motion.

B. STABILITY PERFORMANCE IN CURVED DRIVING

To evaluate the effectiveness of roll motion in stabilizing the robot during curved trajectories, we conducted experiments comparing a TWIP-based motion planner (lacking roll control) with a TWIP-R-based motion planner. Both planners were integrated into the same WBC framework.

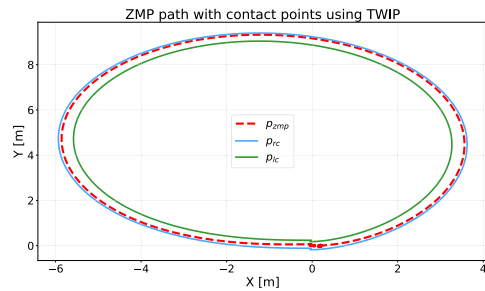


(a)

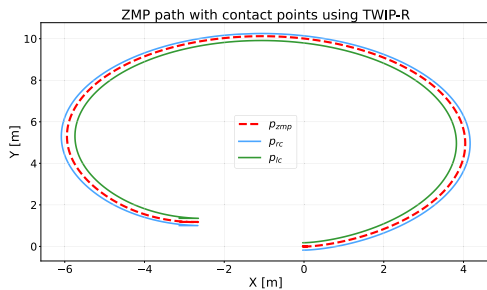


(b)

FIGURE 8. Graph of straight driving: (a) Velocity profiles of the virtual mobile base, showing the desired velocity ($v_{m,d}$) and the actual velocity (v_m). (b) Position and velocity of CoM during motion, illustrating the relationship between the desired and actual trajectories.



(a)



(b)

FIGURE 9. Circular trajectory: (a) ZMP path generated using the TWIP system, showing the trajectory and its relationship with the contact points. (b) ZMP path generated using the TWIP-R system, illustrating the effects of roll control on the ZMP trajectory and contact points.

1) CIRCULAR TRAJECTORY

The robot follows a circular trajectory with steering in one direction, guided by user input specifying a maximum linear velocity of 3.15 m/s and an angular velocity of

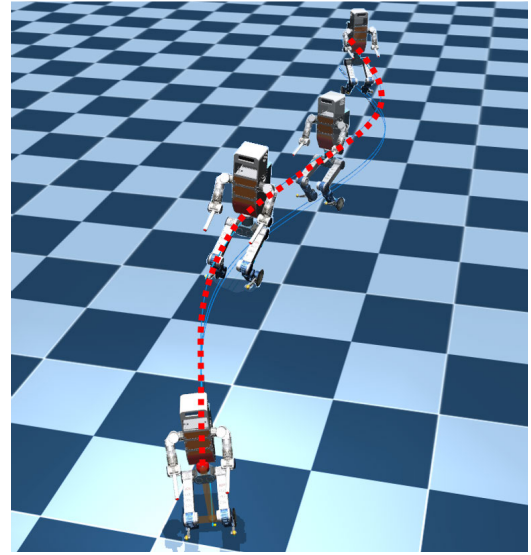
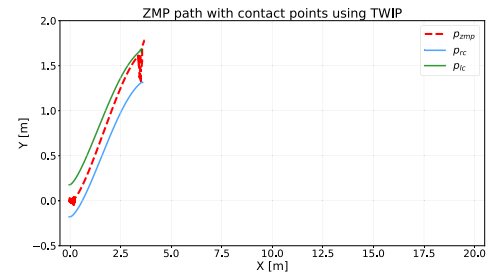
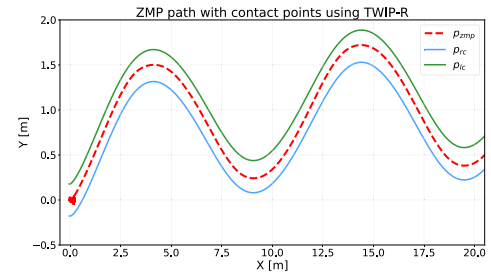


FIGURE 10. A snapshot of the robot during Slalom.



(a)



(b)

FIGURE 11. Slalom trajectory: (a) ZMP path generated using the TWIP system, showing instability as the system fails to shift the ZMP appropriately, resulting in a fall. (b) ZMP path generated using the TWIP-R system, demonstrating stable control through effective roll adjustment during slalom motion.

0.63 rad/s. In Fig. 9(a), the TWIP system, which lacks roll motion control, shows the ZMP shifting toward the outer edge of the ZMP support line during curved motion, increasing instability. In contrast, Fig. 9(b) demonstrates how the TWIP-R system, through roll motion control, attempts to move the ZMP slightly closer to the inside of the ZMP support line, thereby improving stability. This highlights the effectiveness of roll motion in enhancing balance during curved trajectories.

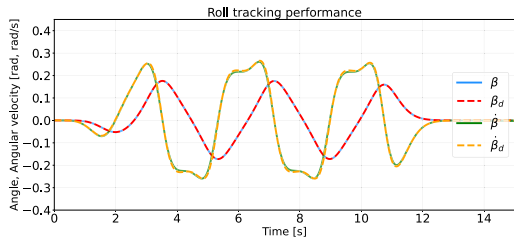
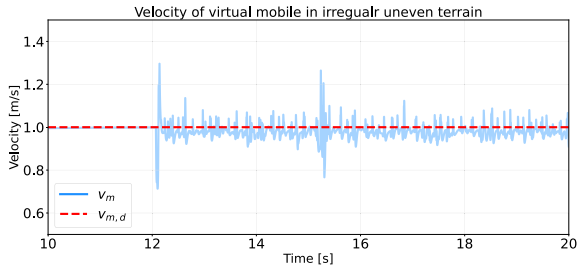
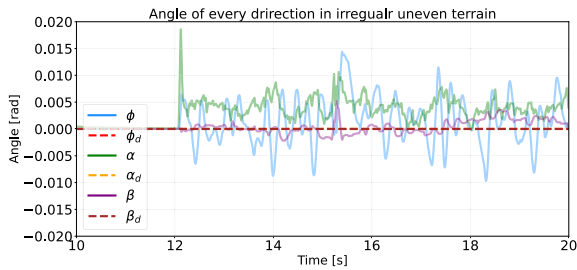


FIGURE 12. Roll motion of slalom driving motion. The leaning angle(β) can be effectively controlled by LQR, managing the CoM motion in the lateral direction.



(a)

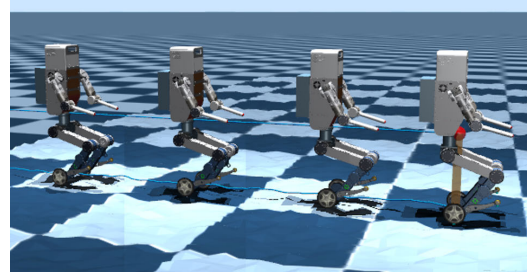


(b)

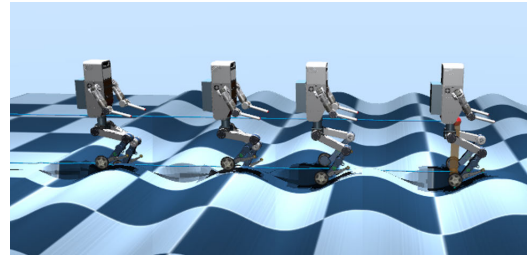
FIGURE 13. Stability in irregular uneven terrain: (a) Tracking velocity performance. (b) To maintain balance, the system aims to keep the pitch, yaw, and roll angles close to zero.

2) SLALOM TRAJECTORY

A snapshot of the robot's slalom simulation is presented in Fig. 10, showcasing its dynamic adaptability. Slalom, which requires rapid and sharp turns at high linear velocities to navigate between obstacles, poses significant challenges for robots employing traditional TWIP systems. Due to the inability of TWIP systems to control roll motion, the ZMP frequently shifts outside the ZMP support line during such maneuvers, leading to instability or falls, as shown in Fig. 11(a). In contrast, the wheeled-legged humanoid robot with the TWIP-R model successfully completes the slalom trajectory while maintaining ground contact. Unlike the TWIP, the TWIP-R utilizes roll motion control to keep the ZMP closer to the center of the ZMP support line, even during high-speed slalom maneuvers with a maximum linear velocity of 3 m/s and an angular velocity of 1.2 rad/s, as shown in Fig. 11(b). The precise control of the roll angle, shown in Fig. 12, highlights the ability of the TWIP-R to effectively manage the CoM motion, ensuring stability throughout the slalom motion.



(a)



(b)

FIGURE 14. Unknown terrain in simulation: (a) Irregular uneven terrain. (b) Wave-like undulations.

C. CONTROL PERFORMANCE IN UNKNOWN ENVIRONMENTS

To evaluate the control stability under unknown conditions, we conducted straight driving simulations on uneven terrain, featuring both irregular ground and wave-like undulations. Fig. 13(a) shows the velocity tracking performance, demonstrating that the robot effectively follows the user-defined input velocity despite surface-induced fluctuations. Additionally, Fig. 13(b) illustrates how the robot maintains its balance by keeping the pitch, yaw, and roll angles within a stable range, despite the external disturbances. For terrain with wave-like undulations, where the two wheels encounter out-of-phase surfaces, the robot exhibits similar stability. Fig. 14 shows the robot's successful navigation across both irregular and wave-like terrains. These results confirm that the ability of the proposed control framework to maintain stability and adhere to the desired trajectory across various terrain conditions.

V. CONCLUSION

In this paper, we propose an optimal control framework integrating a TWIP-R-based motion planner with LQR and a whole-body control approach utilizing QP. The TWIP-R-based motion planner stabilizes the ZMP through roll control, enabling high-speed and sharp-turn maneuvers that surpass the capabilities of conventional TWIP systems. Meanwhile, the WBC ensures the stable operation of the wheeled-legged humanoid robot by effectively coordinating tasks across the entire system, while adhering to physical and dynamic constraints. This allows precise tracking of the dynamic motions generated by the TWIP-R model. The effectiveness of the proposed framework was demonstrated

through various challenging maneuvers, including slalom and curved trajectories, as well as navigation across uneven terrains. These experiments showcased the potential for dynamic and stable locomotion of wheeled-legged humanoid robots.

For future work, we aim to validate the control framework under real-world environments and extend its capabilities to support more dynamic motions, such as jumping and stepping. By actively adjusting contact states, these advanced maneuvers will be integrated into the driving framework, facilitating smooth transitions and simultaneous execution of diverse maneuvers.

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